TrES-2b

R-0116

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_170522` · created 2026-07-16T04:33:54+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457895.79233 +/- 0.00042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1256 +/- 0.0094
Transit depth (Rp/Rs)^2	1.58 +/- 0.24 [%]
Orbital Inclination (inc)	88.86 +/- 0.43
Airmass coefficient 1 (a1)	1.373 +/- 0.012
Airmass coefficient 2 (a2)	-0.1632 +/- 0.004
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	None
Optimal Aperture	4.53
Optimal Annulus	16.41
Transit Duration (day)	0.10915 +/- 0.00078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1256 ± 0.0094 vs 0.1254 (deviation 0.02σ)
Duration fit vs published	0.10915 d vs 0.0746 d (deviation 31.25σ)
Mid-time O–C	-1.3 min
Depth SNR	9.4
Residual scatter	0.88 %

Light curve

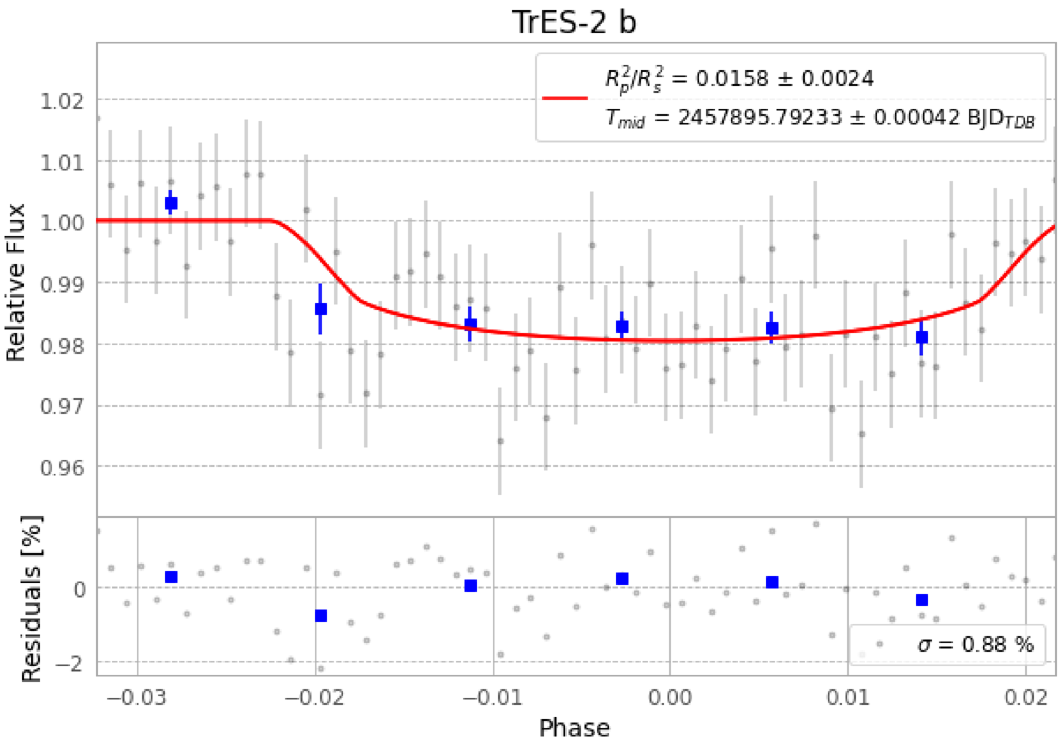

Figure 1 — FinalLightCurve_TrES-2 b_2017-05-21.png (SHA-256 d6d6fb3f053998a6…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [202, 212], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 19c8f453b832b7da…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7`

Data provenance

65 FITS frames (43 MB), `TRES-2170522050312.FITS` → `TRES-2170522081512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-170522.log (334e7a1f8f98…), FinalLightCurve_TrES-2 b_2017-05-21.png (d6d6fb3f0539…), FinalLightCurve_TrES-2 b_2017-05-21.csv (f268f56e2ce0…)

Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)